

Quantum peristalsis: Hydrodynamics and transport in space-time driven quantum systems

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1 Introduction

Steady state currents are generally forbidden in closed quantum systems by symmetry. Specifically, microscopic symmetries prevent a certain direction being privileged over others. In more simple terms, without a reason for a current to flow in a particular direction, a current cannot flow.

Given a problem with whose microscopic physics does not privilege a certain direction, how then can we drive a current? One mechanism is to consider driving protocols with non-trivial space-time symmetries. These symmetries allow us to privilege certain directions, and hence realise steady state currents.

We consider the hydrodynamics of these problems.

2 Space time modulated diffusion

The diffusion equation. A generic thermalising many body quantum system with a Hamiltonian

$$H_0 = \sum_{\mathbf{x}} h_{\mathbf{x}} \quad (1)$$

where $h_{\mathbf{x}}$ is a local term that acts in the vicinity of the position $\mathbf{x}$. We can define a coarse grained energy density averaged over some volume $V(\mathbf{x})$ encapsulating the neighbourhood of $\mathbf{x}$

$$\epsilon(\mathbf{x}, t) = \sum_{\mathbf{x}' \in V(\mathbf{x})} \frac{\langle h_{\mathbf{x}'}(t) \rangle}{V(\mathbf{x}')} \quad (2)$$

Here $V(\mathbf{x})$ is much smaller than the system size, and much larger than microscopic scales. Energy flows into each volume with a rate that is set by Fick's law

$$\partial_t \epsilon = -\nabla \cdot \mathbf{j}_{\epsilon}, \quad \mathbf{j}_{\epsilon} = -D_0 \nabla \epsilon \quad (3)$$

Provided the system is sufficiently microscopically homogenous D_0 is a simple constant, and we arrive at the diffusion equation

$$\partial_t \epsilon = D_0 \nabla^2 \epsilon \quad (4)$$

The solutions of this equation are well known and corresponding to the diffusive spreading of energy through the system, and equilibration to the spatially uniform steady state. Any non-zero currents in the system exist only as vestiges of non-uniformities in the initial state, and decay to zero as the system equilibrates

$$\mathbf{j}_\epsilon \xrightarrow{t \rightarrow \infty} 0 \quad (5)$$

Suppose now we consider the driven cousin of this system

$$H_1 = \sum_{\mathbf{x}} f(\mathbf{x}, t) h_{\mathbf{x}} \quad (6)$$

This is very general family of problems. In the present manuscript we focus on the case where the driving has a continuous space time symmetry

$$f(\mathbf{x}, t) = f(\mathbf{k} \cdot \mathbf{x} - \omega t). \quad (7)$$

If we assume the length scale of this modulation is long $k^{-1} \gg$ (microscopic scales), there are two long wavelength effects of this modulation: First, energy is created in regions of increasing f , and annihilated in regions of decreasing f (this is simply caused by the energy of the systems microstate changing) this results in a source term

$$\partial_t \epsilon = \epsilon \partial_t \log f - \nabla \cdot \mathbf{j}_\epsilon \quad (8)$$

with the two terms corresponding to energy created/annihilate, and energy transported from other parts of the system.

The second important effect is that dynamics progressing faster in the regions where f is large, and slower in the regions where f is small, this is captured by a local modulation to the diffusion constant

$$\mathbf{j}_\epsilon = -D_1 \nabla \epsilon, \quad D_1 = f D_0 \quad (9)$$

Resulting in a modulated diffusion equation

$$\partial_t \epsilon = \epsilon \partial_t \log f + D_0 \nabla \cdot (f \nabla \epsilon) \quad (10)$$

for which $\epsilon = \text{const.}$ is no longer a solution. Perturbative analysis of this equation for a weak drive ($f = 1 + \delta f$) yields instead solutions of the form $\epsilon(\mathbf{x}, t) = e^{\kappa t} \varepsilon(\mathbf{k} \cdot \mathbf{x} - \omega t)$ with a corresponding (spatially averaged) steady state current of $O(\delta f)^2$

$$\bar{\mathbf{j}}_\epsilon = -\bar{\epsilon} \mathbf{v} \cdot \frac{D_0 \kappa}{v^2} + O(\delta f^2) \quad (11)$$

where $\mathbf{v} = \omega \mathbf{k} / k^2$ is the natural velocity scale introduced by the driving protocol (and hence $\bar{\epsilon} \mathbf{v}$ the natural current scale) and $D_0 \kappa / v^2$ is a positive dimensionless constant.

We discuss some of the features of this result (11)

- *Counter-propagating energy current:* The steady state current runs against the sliding wave modulation, that is $\bar{\mathbf{j}}_\epsilon \cdot \mathbf{v} = -\bar{j}_\epsilon v$. This turns out to be a generic feature of protocols of this form.

- *Heating rate κ is quadratic in δf :* Second, the heating is zero to linear order in δf . This follows as the heating is zero in the undriven system, and is be non-negative for both the protocols $f = 1 + \delta f$, and $f = 1 - \delta f$, as such the leading allowed non zero term is quadratic $O(\delta f^2)$.
- *Energy current in quadratic in δf :* That the spatially averaged current also turns on quadratically is clear from application of Fick's law: specifically, writing $\epsilon = \epsilon_0 + \delta\epsilon$ for constant ϵ_0 (i.e. the solution to (10) for $f = 1$) and $\delta\epsilon = O(\delta f)$

$$\bar{j}_\epsilon = -\frac{D_0}{V} \int_V d\mathbf{x} f \nabla \epsilon = -\frac{D_0}{V} \int_V d\mathbf{x} (\nabla \epsilon_0 + \delta f \nabla \epsilon_0 + \nabla \delta\epsilon + \delta f \nabla \delta\epsilon) \quad (12)$$

where the first two terms in the RHS are zero as $\nabla \epsilon_0 = 0$, the third term is the integral of a total derivative, and the final term is $O(\delta f^2)$ as required.

- *Heating rate to current relationship is f -independent:* Though both the heating and current are necessarily $O(\delta f^2)$, it is however a non trivial result that the relationship between them is (to leading order) independent of the driving protocol (i.e. exhibits no dependence on the functional form of f). This is a feature of the leading $O(\delta f^2)$ contribution to $\bar{j}_\epsilon$ and κ and does not survive to higher order corrections.

- Give a cartoon explanation for why the the current is counter propagating
- Do some numerics to check if the f -independence is robust

A clear limitation of this model is the prediction of continued exponential heating. To avoid this unphysical result we need to include some information about the many-body density of states in the model.

3 Few body problem coupled quantum systems

Consider a system of two coupled subsystems H_A and H_B , and the interactions between them V

$$H = H_0 + V, \quad H_0 = H_A + H_B \quad (13)$$

We will assume that the interactions between them are weak, and further that the number of levels in each system is large, so that they may be characterised by the densities of states $\rho_i(E_i)$ of each H_i , or equivalently by their entropies $S_i(E_i) = \log \rho_i(E_i)$.

3.1 Energy transport

As we want to consider energy tranport between the two subsystems, we consider the energy difference between the two sub-systems

$$\Delta H = H_A - H_B \quad (14)$$

from which an energy current operator is straightforward to define

$$J_{B \rightarrow A} = \frac{1}{2} \partial_t \Delta H = \frac{i}{2} [H, H_A - H_B] = \frac{i}{2} [V, H_A - H_B]. \quad (15)$$

Suppose that the system are prepared in some initial microcanonical state σ_i with energies E_A and E_B . We assume that σ_i is a steady state of H_0 , $[H_0, \sigma_i]$. As the systems are coupled by V , if the inverse temperatures $\beta_i := S'_i(E_i)$, are equal, we expect no net energy transfer between the two systems.

$$\langle J_{B \rightarrow A} \rangle = 0 \quad (\beta_A = \beta_B) \quad (16)$$

If however, the two subsystems are at unequal temperatures, we expect heat to flow from the hotter system to the colder one until they equilibrate. We calculate this rate of heat flow via to leading order via two methods: Fermi's Golden rule (App. A), and linear response theory (App. B) [haven't actually finished this section, but it should work(?)], find that

$$\langle J_{B \rightarrow A} \rangle = K_{B \rightarrow A} \Delta\beta + O(\Delta\beta)^2, \quad \Delta\beta = \beta_A - \beta_B \quad (17)$$

where

$$K_{B \rightarrow A} = \left. \frac{\partial \langle J_{B \rightarrow A} \rangle}{\partial \Delta\beta} \right|_{\Delta\beta=0} = \frac{1}{2} \int dt \langle J(t) J(0) \rangle_E \quad (18)$$

where the energy current operator is $J = \frac{i}{2} [V, H_A - H_B]$, and $\langle \cdot \rangle_E$ denotes expectation values with respect to a microcanonical ensemble of total energy $E = E_A + E_B$ with respect to the hamiltonian H_0 .

As a minor wrinkle we note that, some care has to be taken in defining K in a finite system, specifically we want to avoid our attempt to define a transport coefficient from becoming clouded by irrelevant details pertaining to discrete quantisation effects, and the quasiperiodic nature of dynamics in finite quantum systems. In precise terms, by the integral $\int dt C(t)$ in (18), we really mean $\int dt w(t/\tau) C(t)$ where $w(x)$ is some smooth square integrable window function such that $w(0) = 1$, $\int dx w(x) = 1$, and we require the order of limits $\rho(E) \gg \tau \gg (\text{bandwidth of } H_0)^{-1}$ where $\rho(E)$ is the joint density of states. We are then at liberty to take the limit of large systems $\rho(E) \rightarrow \infty$ followed by the limit of $\tau \rightarrow \infty$, recovering (18).

3.2 Driven few body systems

4 Many body coupled quantum systems

From this result, in a coarse grained picture of a many body system we thus expect

$$\mathbf{j}_\epsilon = k(\epsilon) \nabla \beta \quad (19)$$

where the local inverse temperature is defined by the local entropy density

$$\beta = \partial_\epsilon s \quad (20)$$

For an undriven system the energy transport dynamics is then determined by the coupled equations

$$\partial_t \epsilon = -\nabla \cdot \mathbf{j}_\epsilon, \quad \mathbf{j}_\epsilon = k(\epsilon) \nabla \beta, \quad \beta = \partial_\epsilon s \quad (21)$$

where in a generic many body quantum system the density of states will be Gaussian

$$s = s_0 - \frac{1}{2} \log(2\pi w^2) - \frac{1}{2} \left(\frac{\epsilon - \epsilon_0}{w} \right)^2 \quad (22)$$

where ϵ_0 is a local offset, and w is set by the local bandwidth¹, and s_0 is a constant independent of w and ϵ_0 . For an undriven system with no explicit time dependence in the entropy function $\partial_t s = 0$, we find

$$\partial_t \epsilon = -\nabla \cdot (k(\epsilon) \partial_\epsilon^2 s \nabla \epsilon) \quad (24)$$

If ϵ has only negligible deviations from the mean value $\bar{\epsilon}$ this corresponds to diffusion with $D_0 = k(\bar{\epsilon})/w^2$

¹Specifically, at infinite temperature, the energy variance of a box of size V is given by

$$\text{Var}(E_V) = \text{Var} \left(\int_V d^d x \epsilon(x) \right) = V w^2 \quad (23)$$

Appenices

A Fermi Golden Rule derivation of $K_{B \rightarrow A}$

The rate at which the probability that the system is in the initial state (E_A, E_B) decays with a rate set by Fermi's Golden rule

$$\Gamma_{\mu \rightarrow \nu} = 2\pi |V_{\mu\nu}|^2 \rho(E_\nu) \quad (25)$$

where $V_{\mu\nu}$ are the matrix elements of V written in the eigenbasis of H_0 .

To make further progress we need a model for $V_{\mu\nu}$ we use the Schmidt decomposition to write

$$V = \sum_i A_i \otimes B_i. \quad (26)$$

We then assume that A_i satisfies ETH in the eigenbasis of H_A

$$A_{i,ab} = \bar{A}_i(E_a) \delta_{ab} + \sqrt{\frac{\tilde{A}_i(E_a, E_b - E_a)}{\rho_A(E_b)}} R_{ab} \quad (27)$$

Here $\bar{A}_i$ is the *microcanonical expectation function* and $\tilde{A}_i$ is the *spectral function*, the R_{ab} are Gaussian distributed numbers, with unit variance, and uncorrelated with each other. Both $\bar{A}_i$ and $\tilde{A}_i$ are smooth real functions of their arguments, and precisely are defined by

$$\langle A_i \rangle_E = \bar{A}_i(E) \quad (28a)$$

$$\langle A_i(t) A_i(0) \rangle_E - \langle A_i \rangle_E^2 = \int d\omega \tilde{A}_i(E, \omega) e^{i\omega t} \quad (28b)$$

where $\langle \cdot \rangle_E$ denotes the expectation wrt to a microcanonical ensemble defined on subsystem A with energy E . We note that hermittivity requires

$$\tilde{A}_i(E, \omega) \rho_A(E) = \tilde{A}_i(E + \omega, -\omega) \rho_A(E + \omega), \quad (29)$$

We make the corresponding ETH assumption for the matrix elements of B_i .

From (27) we find the RMS off-diagonal matrix elements

$$\begin{aligned} \overline{|\langle E_{A,a}, E_{B,c} | V | E_{A,b}, E_{B,d} \rangle|^2} &= \overline{\left| \sum_i A_{i,ab} B_{i,cd} \right|^2} \\ &= \sum_i \frac{\tilde{A}_i(E_{A,a}, E_{A,b} - E_{A,a})}{\rho_A(E_{A,b})} \frac{\tilde{B}_i(E_{B,c}, E_{B,d} - E_{B,c})}{\rho_B(E_{B,d})} \end{aligned} \quad (30)$$

This for a decay $(E_A, E_B) \rightarrow (E_A + \Delta, E_B - \Delta)$ we take this matrix element, set $(E_{1,a}, E_{2,c}) = (E_A, E_B)$ and $(E_{1,b}, E_{2,d}) = (E_A + \Delta, E_B - \Delta)$ and multiply it by the target density of states $\rho_A(E_{1,b}) \rho_B(E_{2,d})$

$$\Gamma(E_A, E_B) = \int d\Delta \Gamma(E_A, E_B, \Delta), \quad \Gamma(E_A, E_B, \Delta) = 2\pi \sum_i \tilde{A}_i(E_A, \Delta) \tilde{B}_i(E_B, -\Delta) \quad (31)$$

where $\Gamma(E_A, E_B)$ is the decay of the population of the state (E_A, E_B) , and $\Gamma(E_A, E_B, \Delta)$ is the contribution of this overall decay rate from the specific channel $(E_A, E_B) \rightarrow (E_A + \Delta, E_B - \Delta)$.

From this it is natural to define the distribution function $p(E_A)$ which is the probability of measuring E_A energy in the A subsystem, we assume an overall energy $E_T = E_A + E_B$. This distribution function then evolves as

$$\partial_t p(E_A) = \int dE'_A k(E_A, E'_A) p(E'_A) \quad (32)$$

where the transition rates are given by

$$k(E_A, E'_A) = \Gamma(E'_A, E_T - E'_A, E_A - E'_A) - \delta(E_A - E'_A) \Gamma(E'_A, E_T - E'_A) \quad (33)$$

which satisfies probability conservation as required

$$\int dE_A k(E_A, E'_A) = 0 \quad (34)$$

It is straightforward to solve for the steady state of this equation. Specifically the ratio of forward and backward transition rates is given by

$$\frac{k(E_A, E'_A)}{k(E'_A, E_A)} = \frac{p_{\text{eq}}(E_A)}{p_{\text{eq}}(E'_A)} \quad (35)$$

with

$$p_{\text{eq}}(E_A) = \frac{\rho_A(E_A) \rho_B(E_T - E_A)}{\int dE' \rho_A(E') \rho_B(E_T - E')}. \quad (36)$$

It is straightforward to determine that this equilibrium indeed corresponds to the case where the two baths are at the same temperature. Specifically $p_{\text{eq}}(E_A)$ is sharply peaked around the $E_A^{(\text{eq})}$, as defined by

$$\partial_{E_A} \log p_{\text{eq}}(E_A^{(\text{eq})}) = 0 \quad (37)$$

or equivalently, by evaluating (37),

$$S'_A(E_A) - S'(E_T - E_A) = 0, \quad \implies \quad \beta_A = \beta_B. \quad (38)$$

where $\beta_A = S'_A(E_A^{(\text{eq})})$.

A non equilibrium distribution is obtained by applying a Lagrange multiplier to bias the equilibrium distribution

$$p(E_A) = e^{\Delta\beta E_A} p_{\text{eq}}(E_A) \quad (39)$$

where $\Delta\beta$ is the inverse temperature difference between the two systems. This statement can be verified by the same approach

$$\partial_{E_A} \log p(E_A) = \Delta\beta + S'_A(E_A) - S'_B(E_T - E_A) \implies \beta_B = \beta_A + \Delta\beta \quad (40)$$

The rate of energy transfer between the two systems is then given by

$$J = \int dE_A \int dE'_A (E_A - E'_A) k(E_A, E'_A) p(E'_A) \quad (41)$$

at the linear response level this simplifies to

$$J = K \Delta \beta + O(\Delta \beta^2) \quad (42)$$

where

$$\begin{aligned} K &= \left. \frac{\partial J}{\partial \Delta \beta} \right|_{\Delta \beta=0} \\ &= \int dE \int dE' (E - E') k(E, E') E' p_{\text{eq}}(E') \\ &= \frac{1}{2} \int dE \int d\Delta \Delta^2 k(\Delta + E, E) p_{\text{eq}}(E) \end{aligned} \quad (43)$$

Plugging back in the definition of k we obtain

$$\begin{aligned} K &= \frac{1}{2} \int dE \int d\Delta \Delta^2 \Gamma(E, E_T - E, \Delta) p_{\text{eq}}(E) \\ &= \frac{\pi}{N} \sum_i \int dE_A \int d\Delta \Delta^2 \tilde{A}_i(E_A, \Delta) \tilde{B}_i(E_T - E_A, -\Delta) \rho_A(E_A) \rho_B(E_T - E_A) \end{aligned} \quad (44)$$

where $N = \int dE' \rho_A(E') \rho_B(E_T - E') = \rho(E_T)$ is a normalisation constant.

$$\begin{aligned} K &= \frac{\pi}{N} \sum_i \int dE_A \int dE'_A \int dE_B \int dE'_B \delta(E_T - E_A - E_B) \delta(E_T - E'_A - E'_B) \\ &\quad \times (E'_A - E_A)^2 \tilde{A}_i(E_A, E'_A - E_A) \tilde{B}_i(E_B, E'_B - E_B) \rho_A(E_A) \rho_B(E_B) \end{aligned} \quad (45)$$

We then replace the delta functions with nascent deltas of width η , ($\delta \rightarrow \delta_\eta$), and use the identity $\sum_a f(E_{A,a}) = \int dE_A \rho(E_A) f(E_A)$.

$$\begin{aligned} K &= \frac{\pi}{N} \sum_i \sum_{abcd} \delta_\eta(E_T - E_{A,a} - E_{B,c}) \delta_\eta(E_T - E_{A,b} - E_{B,d}) \\ &\quad \times (E_{A,a} - E_{A,b})^2 \frac{\tilde{A}_i(E_{A,a}, E_{A,b} - E_{A,a})}{\rho_A(E_{A,b})} \frac{\tilde{B}_i(E_{B,c}, E_{B,d} - E_{B,c})}{\rho_B(E_{B,d})} \\ &= \frac{\pi}{N} \sum_{abcd} \delta_\eta(E_T - E_{A,a} - E_{B,c}) \delta_\eta(E_T - E_{A,b} - E_{B,d}) \\ &\quad \times (E_{A,a} - E_{A,b})^2 |\overline{\langle E_{A,a}, E_{B,c} | V | E_{A,b}, E_{B,d} \rangle}|^2 \\ &= \frac{\pi}{4N} \sum_{abcd} \delta_\eta(E_T - E_{A,a} - E_{B,c}) \delta_\eta(E_T - E_{A,b} - E_{B,d}) |\overline{\langle E_{A,a}, E_{B,c} | [\Delta H, V] | E_{A,b}, E_{B,d} \rangle}|^2 \\ &= \frac{\pi}{N} \sum_{\alpha\beta} \delta_\eta(E_T - E_\alpha) \delta_\eta(E_T - E_\beta) |\overline{X_{\alpha\beta}}|^2 \end{aligned} \quad (46)$$

where in the last line we have defined $X = \frac{i}{2}[\Delta H, V]$, and $E_\alpha = E_{A,a} + E_{B,c}$ for $\alpha = (a, c)$. We write this as

$$K = \frac{\pi}{\rho(E_T)} \text{tr} [\delta_\eta(E_T - H_0) X \delta_\eta(E_T - H_0) X] \quad (47)$$

and use that $\delta_\eta(E_T - H_0)/\rho(E_T)$ is a microcanonical ensemble ρ_{E_T} at energy E_T where we have used that

$$\text{tr} [\delta_\eta(E_T - H_0)] = \sum_\alpha \delta_\eta(E_T - E_\alpha) = \int dE \rho(E) \delta_\eta(E_T - E) = \rho(E_T). \quad (48)$$

Thus, writing $\omega_{\alpha\beta} = E_\alpha - E_\beta$,

$$K = \pi \text{tr} [X \delta_\eta(E_T - H_0) X \rho_{E_T}] = \pi \sum_{\alpha\beta} \delta_\eta(\omega_{\alpha\beta}) |X_{\alpha\beta}|^2 \rho_{E_T, \alpha\alpha} \quad (49)$$

For the final nascent delta we use the specific form

$$\delta_\eta(\omega) = \frac{1}{2\pi} \int_{-1/\eta}^{1/\eta} dt e^{i\omega t} \quad (50)$$

thus we have

$$K = \frac{1}{2} \int_{-1/\eta}^{1/\eta} dt \sum_{\alpha\beta} e^{-i\omega_{\alpha\beta} t} |X_{\alpha\beta}|^2 \rho_{E_T, \alpha\alpha} = \frac{1}{2} \int_{-1/\eta}^{1/\eta} dt \langle X(t) X(0) \rangle_{E_T} \quad (51)$$

and in the limit $\eta \rightarrow 0$

$$K = \frac{1}{2} \int dt \langle X(t) X(0) \rangle_{E_T} \quad (52)$$

where we note that we require $\eta \rho(E_T) \gg 1$.

B Linear response theory derivation of $K_{B \rightarrow A}$

Section unfinished

The energy difference between the two systems is set by the expectation value of the operator

$$\Delta E = \langle \Delta H \rangle, \quad \Delta H = H_A - H_B \quad (53)$$

The energy current between them is thus set by the rate of change of the operator

$$J_E(t) = \frac{1}{2} \partial_t \Delta E(t) = \langle \partial_t \Delta H(t) \rangle = \frac{i}{2} \langle [H, \Delta H(t)] \rangle \quad (54)$$

We evaluate this quantity with respect to the ensemble

$$\rho = \frac{1}{Z} \exp [-\beta H - \Delta \beta \Delta H] \quad (55)$$

where $H = H_A + H_B + V$, and we have introduced the Lagrange multiplier $\Delta\beta$, coupled to ΔH in order to probe the behaviour away from equilibrium. As $[\Delta H, H_A] = [\Delta H, H_B] = 0$ it is useful to adopt a notation

$$\beta H + \Delta\beta \Delta H = \beta(H_0 + V') \quad (56)$$

where V' is the off diagonal part of V or explicitly

$$H_0 = \left(1 + \frac{\Delta\beta}{\beta}\right) H_A + \left(1 - \frac{\Delta\beta}{\beta}\right) H_B + V_0. \quad (57)$$

where V_0 is the diagonal part of V .

Expanding in powers of V we have

$$\rho = \rho_0 + \int_0^\beta d\tau V'(\tau) \rho_0 + O(V'^2) \quad (58)$$

where $\rho_0 = \rho|_{V=0} = \exp[-\beta H_0 - \Delta\beta \Delta H]/Z_0$, and $H_0 = H|_{V=0} = H_A + H_B$ and $V(\tau)$ is the usual imaginary time evolved operator

$$V(\tau) = e^{-\tau H_0} V e^{\tau H_0} \quad (59)$$

Plugging this into J_E we have

$$\begin{aligned} J_E &= i(1 - \beta \langle V \rangle_0) \langle [V, \Delta H] \rangle_0 + i \int_0^\beta d\tau \langle [V, \Delta H] V(\tau) \rangle_0 + O(V^3) \\ &= i \int_0^\beta d\tau \langle [V, \Delta H] V(\tau) \rangle_0 + O(V^3) \end{aligned} \quad (60)$$

where we have used that $[H_0, \Delta H] = 0$. Expanding this out this first term evaluates to zero, whereas the second term is given by

$$J_E = i \text{tr} [] \quad (61)$$

A Perturbative analysis of Eq. (10)

Taking (10), substituting the ansatz $\epsilon(\mathbf{x}, t) = e^{\kappa t} \epsilon(\mathbf{k} \cdot \mathbf{x} - \omega t)$ with $\epsilon(z) = \epsilon(z + 2\pi)$, and making the replacements $\tilde{D} = D_0 k/v$ (with $v = \omega/k$) and $\lambda = \kappa/\omega$ yields the eigenvalue equation

$$\Lambda[\epsilon(z)] = \lambda \epsilon(z) \quad (62)$$

for the linear operator

$$\Lambda[\epsilon] = \tilde{D} \partial_z (f \partial_z \epsilon) + \partial_z \epsilon - \epsilon \partial_z \log f \quad (63)$$

where the long time dynamics is set by the solution with the largest eigenvalue λ (the fastest growing instability). Expanding $f(z) = 1 + \eta f_1(z)$ to leading order in η , $\Lambda = \sum_{n=0}^{\infty} \eta^n \Lambda_n$

$$\begin{aligned} \Lambda_0[\epsilon] &= \tilde{D} \partial_z^2 \epsilon + \partial_z \epsilon, \\ \Lambda_1[\epsilon] &= \tilde{D} \partial_z (f_1 \partial_z \epsilon) - \epsilon \partial_z f_1 \\ \Lambda_{n \geq 2}[\epsilon] &= -\epsilon (-f_1)^{n-1} \partial_z f_1 \end{aligned} \quad (64)$$

at zeroth order in η the eigenvalue equation reads

$$\Lambda_0[\epsilon_0] = \lambda_0 \epsilon_0 \quad \implies \quad \lambda_0 = 0, \quad \epsilon_0 = \text{const.} \quad (65)$$

whereas at first order we have

$$\Lambda_0[\epsilon_1] + \Lambda_1[\epsilon_0] = \lambda_0 \epsilon_1 + \lambda_1 \epsilon_0. \quad (66)$$

Explicitly this becomes

$$\tilde{D} \partial_z^2 \epsilon_1 + \partial_z \epsilon_1 - \epsilon_0 \partial_z f_1 = \lambda_1 \epsilon_0 \quad (67)$$

with the solution $\lambda_1 = 0$, and

$$\begin{aligned} \epsilon_1(z) &= \frac{\epsilon_0}{\tilde{D}} \int_0^\infty dz' e^{-z'/\tilde{D}} f_1(z - z') \\ &= \frac{\epsilon_0}{\tilde{D}} \sum_{n=0}^{\infty} e^{-2\pi n/\tilde{D}} \int_0^{2\pi} dz' e^{-z'/\tilde{D}} f_1(z - z') \\ &= \frac{\epsilon_0}{\tilde{D}} \cdot \frac{1}{1 - e^{-2\pi/\tilde{D}}} \cdot \int_0^{2\pi} dz' e^{-z'/\tilde{D}} f_1(z - z') \end{aligned} \quad (68)$$

The second order term is fixed by

$$\begin{aligned} \lambda_2 \epsilon_0 &= \Lambda_0[\epsilon_2] + \Lambda_1[\epsilon_1] + \Lambda_2[\epsilon_0] \\ &= \tilde{D} \partial_z^2 \epsilon_2 + \partial_z \epsilon_2 + \tilde{D} \partial_z f_1 \partial_z \epsilon_1 + \tilde{D} f_1 \partial_z^2 \epsilon_1 - \epsilon_1 \partial_z f_1 + \epsilon_0 f_1 \partial_z f_1 \\ &= \tilde{D} \partial_z^2 \epsilon_2 + \partial_z \epsilon_2 + \tilde{D} \partial_z f_1 \partial_z \epsilon_1 - \partial_z (f_1 \epsilon_1) + \epsilon_0 \partial_z f_1^2 \end{aligned} \quad (69)$$

integrating both sides of this equation we kill many of the terms as they are total derivatives

$$\lambda_2 \varepsilon_0 = \frac{\tilde{D}}{2\pi} \int_0^{2\pi} dz \partial_z f_1 \partial_z \varepsilon_1 \quad (70)$$

which by substitution yields

$$\lambda_2 = \frac{1}{2\pi} \cdot \frac{1}{1 - e^{-2\pi/\tilde{D}}} \int_0^{2\pi} dz \int_0^{2\pi} dz' e^{-z'/\tilde{D}} f_1'(z) f_1'(z - z') \quad (71)$$

By substituting $f_1(z) = \sum_{q \in \mathbb{Z}} f_{1,q} e^{iqz}$ the integrals can be evaluated and yield

$$\lambda_2 = \sum_{q \in \mathbb{Z}} |f_{1,q}|^2 \frac{i\tilde{D}q^2}{i + \tilde{D}q} = 2 \sum_{q=1}^{\infty} |f_{1,q}|^2 \frac{\tilde{D}q^2}{1 + (\tilde{D}q)^2} \quad (72)$$

Yielding the overall heating rate

$$\kappa = \lambda\omega = 2\omega\eta^2 \sum_{q=1}^{\infty} |f_{1,q}|^2 \frac{\tilde{D}q^2}{1 + (\tilde{D}q)^2} + O(\eta^3) \quad (73)$$

A.1 Spatially averaged current

We now calculate the spatially averaged current

$$\begin{aligned} \bar{j}_\epsilon &= \frac{1}{V} \int_V d\mathbf{x} j_\epsilon \\ &= -\frac{D_0}{V} \int_V d\mathbf{x} f(\mathbf{x}, t) \nabla \epsilon(\mathbf{x}, t) \\ &= -\bar{\epsilon} \mathbf{v} \cdot \frac{\eta^2 \tilde{D}}{2\pi \varepsilon_0} \int_0^{2\pi} dz f_1(z) \varepsilon_1'(z) + O(\eta^3) \\ &= -\bar{\epsilon} \mathbf{v} \cdot \frac{\eta^2}{2\pi} \cdot \frac{1}{1 - e^{-2\pi/\tilde{D}}} \int_0^{2\pi} dz \int_0^{2\pi} dz' e^{-z'/\tilde{D}} f_1(z) f_1'(z - z') + O(\eta^3) \end{aligned} \quad (74)$$

where $\bar{\epsilon} = e^{t/\tau} \varepsilon_0$ is the mean energy density.

We now show that the mean current always flows in the opposite direction to $\mathbf{v}$. By substituting $f_1(z) = \sum_{q \in \mathbb{Z}} f_{1,q} e^{iqz}$ for real f (i.e. $f_{1,-q} = f_{1,q}^*$) the integral can be evaluated and yields

$$\bar{j}_\epsilon = -\bar{\epsilon} \mathbf{v} \cdot 2\eta^2 \sum_{q=1}^{\infty} |f_{1,q}|^2 \frac{(\tilde{D}q)^2}{1 + (\tilde{D}q)^2} + O(\eta^3) \quad (75)$$

as all the terms in this are manifestly positive, the overall direction of the current is set by the minus sign, and hence flows against $\mathbf{v}$.

Comparing this with our previous result for the heating rate, we obtain

$$\bar{j}_\epsilon = -\bar{\epsilon} \mathbf{v} \frac{D_0 \kappa}{v^2} + O(\eta) \quad (76)$$

as the system always heats ($\kappa > 0$) the current directions always opposes the sliding direction of the driving protocol.